



ClimaSentinel

Live Climate Risk Dashboard & Automated Climate Pipeline



CLIMASENTINEL

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Context & Mission

Protecting enterprise operations from climate volatility.

The Enterprise Challenge: Extreme weather variability directly threatens physical assets and business continuity, heatwaves trigger data center cooling failures, floods disrupt supply chains, and severe storms halt transit networks. Standard weather apps do not provide operational risk alerts.

The ClimaSentinel Solution: We transform daily weather, air-quality, river, and climate-baseline signals into a proactive **Global Tipping Risk Score** across 10 major European hubs. Designed for decision makers, it delivers live situational awareness to prevent operational downtime, protect infrastructure, and trigger safety protocols.

10

European hubs

LIVE

risk dashboard

0

marginal cloud cost

Objectives: Business Impact First

What decision makers should remember

- Live situational awareness:** a city can be understood in seconds instead of reading multiple weather, pollution and flood feeds.
- One score, many signals:** heat, wind, air quality, river discharge and climate baselines become one simple city-level risk story.
- Decision-ready outputs:** every city receives a risk level, primary threat and plain-language explanation rather than raw technical tables.
- Automated economics:** scheduled cloud ingestion keeps the dashboard lightweight and repeatable.

User experience

- GREEN safe YELLOW watch RED act
- Premium climate-tech interface with color-coded risks, glowing threat indicators and refined progress bars.

Citizen-friendly Recruiter-ready Policy-oriented Production-minded

Method: Live Climate Intelligence Pipeline

A clean 3-layer climate data factory

1. Collection

BRONZE

Every morning at 06:00 UTC, the system fetches fresh weather, air-quality, flood, ERA5 and CMIP6 data for each city.

2. Refinement

SILVER

Raw feeds are cleaned, deduplicated and standardized in BigQuery so the dashboard uses trusted city signals.

3. Live Insights

GOLD

Business tables produce current score, risk color, dominant threat and rankings for the live dashboard.

Dashboard-first intelligence

- Current climate status:** live view of today’s risk level, main threat and city comparison.
- Experimental forecast:** 7-day Random Forest tracked with MLflow — a research extension, not the core promise.

Raw APIs → Trusted Data → Live Dashboard

Results: Live Dashboard for Climate Now

What decision makers understand in seconds

Status

Is my city safe?

Green / Yellow / Red

Driver

Why is risk rising?

Heat, air, wind, river

Compare

Where to focus?

10 cities ranked

ClimaSentinel

LIVE CLIMATE INTELLIGENCE

Global Tipping Risk

Real-time monitoring of critical climate thresholds and driver factors across major European metropolises.

ACTIVE REGIONS

10

AVG GLOBAL RISK

55.4

HIGHEST RISK REGION

PARIS, FR

PARIS, FR

DRIVER: HEAT

TIPPING SCORE

83.8

LONDON, GB

DRIVER: HEAT

TIPPING SCORE

74.0

AMSTERDAM, NL

DRIVER: HEAT

TIPPING SCORE

69.5

LIVE

current climate

10

city views

06:00

daily refresh

Conclusion: Why ClimaSentinel Matters

Delivering Proven ROI & Enterprise Resilience

- Proactive Risk Mitigation:** Moving enterprises from reactive crisis response to live climate intelligence preventing costly infrastructure downtime and supply chain disruptions.
- Enterprise-Grade Automation:** Powered by a fully automated cloud pipeline that guarantees fresh daily insights with zero manual overhead or maintenance infrastructure costs.
- Strategic Decision Advantage:** Decision makers gain instant situational awareness across all operational hubs, bridging the gap between raw environmental data and confident executive action.

“From raw environmental signals to confident enterprise action.”

Roadmap: Next Evolution

Roadmap for a stronger live tool

Scale coverage

Move from 10 cities to more regions, districts and vulnerable local zones.

Send alerts

Email, SMS or municipal webhooks when a risk threshold is crossed.

Explain the score

Plain-language drivers, confidence wording and clear city summaries for non-technical users.

Validate forecasts

Keep the 7-day model experimental until accuracy, drift and safety gates prove it reliable.

Principle: publish what is reliable now, label predictions honestly, and make every climate signal easy to act on.

KEY REFERENCES

- [1] Open-Meteo: weather, air quality, flood, ERA5 and CMIP6 APIs.
- [2] Google Cloud, BigQuery, dbt, FastAPI, Next.js and monitoring stack.
- [3] ClimaSentinel GitHub repository, source code and project documentation.

TECH STACK

FastAPI

Next.js

Open-Meteo

BigQuery

dbt

MLflow

DVC / DagsHub

Built as a climate intelligence platform for Innovation Project M1 EFREI.

Encadrant: Rachid Taalout

MORE INFORMATION

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